

Bayt Al Hekma (بيت الحكمة) Library Management System

Complete Program Description

A complete description of what the program does, what it represents, how each part works, and why the design is organized this way.

1. What This Program Is

The Bayt Al Hekma Library Management System is a console application for managing a small community library.

It has two main jobs. First, it manages library items, who has borrowed them, their current status, and how long they can be borrowed. Second, it calculates late-return fines, applies the correct member waiver, adds an administrative charge that cannot be waived, and records the amount owed by each member.

The program uses one common parent type for all library items. Each item can decide how long it may be borrowed and how much a late return costs. Books, magazines, and DVDs can give different answers.

The rest of this document explains the classes, rules, and operations.

2. The Building Blocks

This section describes each class, what data it stores, what it can do, and what it should not allow.

3.1 ItemStatus — an enumeration

Use three values:

- **AVAILABLE** — the item can be borrowed.
- **ON_LOAN** — the item is currently borrowed.
- **LOST** — the item is lost and cannot be borrowed.

An item must always have exactly one of these three statuses.

3.2 MembershipType — an enumeration that carries data

Use three membership types. Each type stores its own waiver rate.

This means the program does not need a separate table that connects membership types to percentages. Each enum value knows its own rate. Adding another type should require only a small change in the enum.

3.3 Renewable — an interface

The interface has two operations:

- Renew the loan — try to extend the loan and report whether it worked.
- Report the renewal limit — return the maximum number of renewals allowed.

Books and magazines implement this interface. DVDs do not.

All three classes are library items, so they share the same parent class. Only two types have the ability to renew. The interface is used to represent this ability.

3.4 Member

A Member stores:

- Name
- Membership ID, which does not change
- Membership type, which does not change
- Balance owed
- Number of items currently held

A Member can:

- Report all of its information.
- Change its name.
- Receive a fine.
- Make a payment.
- Check whether it is currently allowed to borrow.
- Record a borrowing.
- Record a return.

The Member class must control its balance and item count. There should be no method that lets another class set these values directly.

- A fine must be positive.
- A payment must be positive and cannot be greater than the balance owed.
- A borrowing increases the item count by 1.
- A return decreases the item count by 1, but the count cannot go below 0.

A member can borrow only when both conditions are true:

- The member currently holds fewer than **3 items**.
- The member owes no more than **100 EGP**.

The class has two constructors: one for a new member and one for a member who already has an outstanding balance. The constructors should avoid repeating the same setup code.

3.5 LibraryItem — an abstract class

LibraryItem is the abstract parent class for all library items.

It stores common information:

- Catalogue ID
- Title
- Current status
- **Borrower Member ID**
- Number of renewals used

The borrower is identified by the member's unique membership ID, not by the member's name.

It also stores library-wide information such as the library name, administrative charge, and the total number of items ever added.

LibraryItem can:

- Report all of its properties.
- Be marked as lost and brought back when appropriate.
- Lend itself to a member only when it is AVAILABLE.
- Be returned. Returning an item resets its status, borrower member ID, and renewal count.
- Record a renewal.

LibraryItem cannot be created directly because it is abstract.

Every child class must provide three answers:

- How much is the fine for a given number of overdue days?
- How many days can this item be borrowed?
- What is the name of this item category?

The display method should be written once in LibraryItem. It should display the ID, category, title, status, borrower member ID, loan period, and one-day fine. The child class supplies the item-specific values.

The return operation should not be overridden by the child classes. Returning an item must always reset it correctly.

3.6 The three item types

Each item type extends `LibraryItem` and provides its own extra fields and rules.

Item	Extra fields	Loan period	Fine	Renewal
Book	Author, page count	14 days	5 EGP / overdue day	2 times
Magazine	Issue number	7 days	3 EGP / overdue day; maximum 30 EGP	1 time
DVD	Runtime	3 days	15 EGP / overdue day	Not allowed

Book: adds an author and page count. Loan period **14 days**. Fine **5 EGP per overdue day**, with no maximum. Renewable up to **2 times**.

Magazine: adds an issue number. Loan period **7 days**. Fine **3 EGP per overdue day**, with a maximum of **30 EGP**. Renewable **once**.

DVD: adds a runtime. Loan period **3 days**. Fine **15 EGP per overdue day**, with no maximum. **Not renewable**.

The magazine has a maximum fine, so its fine rule is different from simply multiplying the daily rate by the number of overdue days.

3.7 Library

The Library holds two collections: the catalogue and the member register. Each collection has a count of how many entries are currently used.

Registering:

- Reject an item if the catalogue is full or its ID already exists.
- Reject a member if the member register is full or the membership ID already exists.

Finding:

- Find an item by catalogue ID.
- Find a member by membership ID.
- If there is no match, return nothing and let the caller handle the result.

Listing:

- List the whole catalogue.
- List items filtered by status.
- List all members and how much each member owes.

Calculations:

- Items on loan — the number of items currently borrowed.
- Loan rate — the percentage of catalogue items currently on loan.
- Total outstanding — the total balance owed by all members.
- Projected fines — if every item currently on loan is returned a given number of days late, calculate the fines before membership waivers.

Lending: Find the item and member. Reject the operation if either is missing, the item is not AVAILABLE, or the member is not eligible. On success, record the member's **membership ID** as the borrower and increase the member's item count.

Returning: Find the item and check that it is ON_LOAN. Reject a negative overdue value. Find the member using the **borrower membership ID** stored in the item before returning it, because returning the item clears that ID. Calculate the fine, charge the member, decrease the member's item count, display the fine breakdown, and return the item.

Renewing: Find the item. Check whether it supports renewal. If it does, try to renew it. The renewal can still fail if the item is not on loan or has reached its limit. If the item does not support renewal, display a clear message.

Marking an item as lost: Find the item by ID. The item can be marked as LOST only when its current status is AVAILABLE. Change its status to LOST. A LOST item cannot be borrowed again. If the item is currently ON_LOAN, reject the operation.

Reporting

The report should show the catalogue size, number of items ever added, number of items on loan, loan rate, total outstanding fines, and the projected fine amount.

3.8 Main

Main creates the library, adds sample items and members, then repeatedly displays the menu, reads the user's choice, performs the selected operation, and continues until the user exits.

Book, Magazine, and DVD objects should all be stored in the same LibraryItem collection. The Library should not need to know the exact item type for normal item operations.

7. Every Operation In Detail

The program presents **11 operations plus Exit**.

No.	Operation	Description
1	View catalogue	No input. Display every item: ID, category, title, status, borrower member ID, loan period, and one-day fine.
2	Register member	Ask for name, membership ID, and category. Reject duplicate IDs or a full register. Start with a zero balance.
3	Borrow item	Ask for item ID and membership ID. Reject missing IDs, an unavailable item, or a member who holds 3 items or owes more than 100.00 EGP.
4	Return item	Ask for item ID and overdue days. Zero means on time. Reject negative days. Calculate the fine, charge it, display the breakdown, and reset the borrower's count.
5	Renew loan	Ask for an item ID. Check whether its type can be renewed and whether its renewal limit has been reached.
6	Search item by ID	Display that item's details, or report that it was not found.
7	View items by status	Ask for AVAILABLE, ON_LOAN, or LOST and display matching items, or report that there are none.
8	Pay outstanding fines	Ask for membership ID and payment amount. Reject non-positive payments and payments greater than the balance.
9	View all members	Display name, ID, category, items held, and balance owed.
10	Library report	Display catalogue size, items ever added, items on loan, loan rate, total outstanding, and projected fines for a 5-day overdue.
11	Mark item as lost	Ask for an item ID. The item must currently be AVAILABLE. Change its status to LOST. A LOST item cannot be borrowed. If the item is already LOST, report an error.
0	Exit	End the program.

8. Every Rule The Program Enforces

Catalogue rules

- Catalogue IDs are unique; a duplicate ID is rejected.
- An item's catalogue ID and title do not change after creation.
- An item can be lent only when its status is AVAILABLE.
- An item with status LOST cannot be lent.
- An item can be marked as LOST only when its current status is AVAILABLE.
- If an item is currently ON_LOAN, an attempt to mark it as LOST is rejected.
- An item can be returned only when its status is ON_LOAN.
- Returning an item always resets its status, borrower member ID, and renewal count.

Member rules

- Membership IDs are unique; a duplicate ID is rejected.
- A membership ID and category do not change.
- A member can hold at most 3 items at one time.
- A member who owes more than 100.00 EGP cannot borrow.
- A charged fine must be positive.
- A payment must be positive and must not be greater than the balance owed.
- The balance can never become negative.

- The number of held items can never become negative.
- The balance and held-item count cannot be set directly.

Fine rules

- Overdue days cannot be negative.
- 0 overdue days means no fine and no administrative charge.
- The magazine fine cannot exceed 30.00 EGP.
- The membership waiver applies only to the base fine, not the administrative charge.
- The member must be identified using the borrower member ID before the item is returned/freed.

Renewal rules

- Only item types that implement Renewable can be renewed.
- An item can be renewed only while it is ON_LOAN.
- An item cannot be renewed beyond its type's limit.
- Trying to renew a type that cannot be renewed must display an explanation instead of crashing.

General rules

- Neither collection can exceed its capacity.
- Searching for an item or member that does not exist must report the problem and never crash.
- Every operation returns to the menu instead of ending the program.